Boundary-Only Borel Constant-Term Complex

Autonomous discussion Pass 95 packages the all-prime Borel boundary shadow as a two-term constant-term complex.

The complex is

$$C_B = \mathbb{Q}^\times \ltimes [\mathbb{Q} \to \mathbb{A}_f],$$

with cohomological degrees $0 \to 1$. The map is the diagonal inclusion $\mathbb{Q} \hookrightarrow \mathbb{A}_f$, and the global Levi $\mathbb{Q}^\times$ acts by scalar multiplication.

Its all-prime solid cohomology is

$$H^0(C_B) = 0, \qquad H^1(C_B) = \mathbb{A}_f/\mathbb{Q} \cong \widehat{\mathbb{Z}}/\mathbb{Z} = \epsilon.$$

Thus C_B carries the Pass-94 boundary-only functional-equation shadow.

At finite conductor N, the shadow is

$$C_{B,N} = (\mathbb{Z}/N)^\times \ltimes \left[\mathbb{Z}/N \to \prod_{p^e \| N} \mathbb{Z}/p^e\right].$$

The diagonal map is an isomorphism by the Chinese remainder theorem, so every fixed finite conductor shadow is ordinary-acyclic. The phantom is not a finite-level cokernel; it is the all-prime solid boundary.

Naturality has two directions. If $N \mid M$, conductor reduction gives a commuting square of two-term complexes and preserves the Borel unit class. For finite supports $S \subseteq T$, projection $T \to S$ is canonical. Support enlargement $S \to T$ is only a finite-conductor CRT choice or span: exact zero-insertion does not preserve the all-prime diagonal copy of $\mathbb{Z}$.

This gives a precise "functional equation without Weyl operator" theorem. The signed boundary and constant term survive, but there is no nontrivial Whittaker coefficient and no standard Weyl/Fourier intertwiner.

The finite checker `artifacts/reports/pass95-boundary-only-borel-constant-term-complex-check.json` verifies finite conductor acyclicity, conductor naturality, support projection behavior, and the all-prime constant-term row with solid boundary $\epsilon = \widehat{\mathbb{Z}}/\mathbb{Z}$.